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23 February 1976
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8 June 1965

MILITARY SPECIFICATION

TESTER, HOSE ASSEMBLIES, HYDRAULIC PNEUMATIC HYDROCARBON FUEL AND OIL AIRCRAFT AND MISSILE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the description of a test unit suitable for conduction proof pressure and leakage tests on aircraft and missile hose assemblies which conform to MIL-H-8788, MIL-H-8790, and MIL-H-8795.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-250/11	Aluminum Alloy 6061, Plate and Sheet
QQ-P-416	Plating, Cadmium (Electrodeposited)
TT-E-489	Enamel, Alkyd, Gloss (for Exterior and Interior Surface)
TT-L-32	Lacquer Cellulose Nitrate, Gloss, for Aircraft Use
TT-P-1757	Primer Coating Zinc Chromate Low Moisture Sensitivity

FSC 4940

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PPP-B-601	Box, Wood, Cleated Plywood Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Packaging, Methods of
MIL-G-5510	Packing; Preformed Straight Thread Tube Fitting Boss
MIL-G-5514	Gland Design Packings, Hydraulic, General Requirements for
MIL-P-5516	Packing, Preformed, Petroleum Hydraulic Fluid Resistant, 160 Deg F
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-P-6808	Primer Coating, Zinc Chromate, Process for Application of
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	Screw Threads, Standard Optimum Selected Series, General Specification for
MIL-H-82382	Hydraulic Fluid, Fire Resistant
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-H-8788	Hose, Hydraulic and Pneumatic, High Pressure
MIL-H-8790	Hose Assemblies, Rubber, Hydraulic, High Pressure (3000)
MIL-H-8795	Hose Assemblies, Rubber, Hydraulic, Fuel and Oil Resistant

STANDARDS

FEDERAL

FED-STD-601	Rubber, Sampling and Testing

MILITARY

MIL-STD-100	Drawings, Engineering and Associated Lists
MIL-STD-129	Marking for Shipment and Storage

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MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Standards and Specification Order of Precedence for the Selection of
MIL-STD-808	Finishes, Protective, and Codes for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods
MS28741	Hose Assembly, Detachable End Fitting, Medium Pressure
MS28759	Hose Assembly, Rubber, Hydraulic and Pneumatic (3000 PSI) Flared Tube
MS28762	Hose Assembly, Hydraulic and Pneumatic High Pressure (3000 PSI) Flareless Tube

AIR FORCE-NAVY AERONAUTICAL

AN6236	Filter Element, Hydraulic Replaceable Micronic Reservoir Type
AND10059	Fitting Styles of Pipe, Threaded, Flared Tube, and Hose

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First Article. Unless otherwise specified (see 6.2), this specification requires a first article, as soon as practicable, after award of contract or order, and tester, hose assemblies, shall be subject to examination and testing to determine conformance with this specification (see 4,2.1 and 6.2). Approval of the first article by the activity concerned shall not relieve the contractor of his responsibility to supply equipment conforming to this specification, Any change or deviation from the approved first article shall be subject to the prior written approval of the contracting officer.

3.2 Components. Components of the tester shall consist of a 5-gallon reservoir, the necessary control valves, fittings, regulators, gages, and an air operated automatic pump.

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3.3 Selection of Specifications and Standards. Specification and Standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3.1 Standard parts. AN or MIL standard parts shall be used wherever they are suitable for the purpose and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.4 Material. All material used shall be as specified herein and as shown on applicable drawings. Material not specified shall be suitable for the purpose intended. The material shall be free from all defects and imperfection that might affect the serviceability of the finished product. All metals shall be of a corrosion resistant type or shall be suitably protected to resist corrosion during the normal service life of the unit.

3.5 Design. All high pressure components, exclusive of gages, shall be capable of withstanding a minimum of 1-1/2 times the maximum operating pressure. The complete tester shall occupy a space not greater than 18 inches by 16 inches by 25 inches.

3.6 performance. The tester shall be capable of boosting the hydraulic pressure at a minimum ratio of 150 (hydraulic) to 1 (air). The unit shall be capable of producing a minimum fluid pressure of 15,000 psi, utilizing compressed air of 100 psi as a power source. The rate of pressure buildup should be at the rates specified in FED-STD-601, Method 10011. The tester shall operate satisfactorily with no indication of external leakage when subjected to the proof pressure test and leakage test.

3.7 Details of components.

3.7.1 Hydraulic components.

3.7.1.1 Adapters. Suitable adapters shall be furnished with each tester to permit connecting hose assemblies MS28741-3 to -16 inclusive, and MS28762-3 to -16 inclusive, to the oil ports. These adapters shall be stored in racks provided on each side of the unit.

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3.7.1.2 Fittings. Wherever possible, those fittings conforming to AND10059 shall have threads in accordance with MIL-S-7742 and MIL-P-7105 as applicable.

3.7.1.3 Filter. A hydraulic ilter utilizing an AN6236A filter element shall be included in the unit between the reservoir and suction side of the pump or within the reservoir. The filter element shall be easily removed for replacement in service.

3.7.1.4 Fluid. The unit shall be designed for use with oil conforming to MIL-H-5606 or MIL-H-82382.

3.7.1.5 Gages. The tester shall be equipped with two hydraulic commercial type pressure gages with a minimum dial diameter of 3½ Inches, and one commercial type air pressure gage with a minimum dial diameter of 2 inches. One of the hydraulic gages shall have a range of 0 to 3000 psi with increments of 25 psi. The second gage shall have a range of 0 to 15,000 psi with increments of 100 psi. The hydraulic gages shall be of the slotted link type to prevent damage to the gages on sudden release of pressure. A gage saver shall be incorporated before the low pressure gage to protect it when pressures exceed the maximum range of the gage, The air pressure gage shall have a range of 0 to 160 psi with increments of 5 psi. The gage cases may be brass, steel, or aluminum.

3.7.1.5.1 Sight gage. Means shall be provided to fill the test hose under low pressure. This shall include provisions for bleeding all air out of the hydraulic test circuit and a sight tube to permit visual inspection of the bleeding operation. The sight tube shall be horizontally mounted and positioned so that it does not siphon air bubbles from the discharge side. The sight tube shall not be subject to operating pressure.

3.7.1.6 Reservoirs. The reservoir shall be fabricated from 6061-T6 aluminum alloy sheet, conforming to QQ-A-250/11. The minimum thickness of the top, sides, and bottom shall be ,062 inch. The reservoir shall have a capacity of 5 gallons plus an expansion volume of at least 1/2 gallon above the full (5 gallon) level, The top of the reservoir shall be of sufficient strength to provide rigid mounting and alignment for components, brackets, et cetera. Angles or legs, of the same material as the reservoir and suitable strength, shall be welded to or fastened at the bottom of the reservoir to support or tie down the complete unit. A minimum of 4 holes, 13/32 diameter or larger, shall be located in the base angles or legs. If the top of the reservoir is not removable, a

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clean-out hole of at least 5 inches in diameter and a drain valve shall be furnished. A commercial type filler cap with micronic filter vent and filler neck provided with a 30 mesh or finer strainer shall be installed for replenishing fluid in the reservoir. The reservoir shall be provided with a fluid level sight gage to indicate full level, Racks for holding and storing hose test fittings shall be attached to the side of the reservoir.

3.7.1.7 Pumps. The air operated pump shall have a minimum oil displacement of 0.20 cubic inch per stroke or cycle and a capacity of 0.30 gpm when operating at approximately 100 psi hydraulic pressure with an inlet pressure of 100 psi. The pump shall not require more than 8 cfm at 100 psig to operate it at any hydraulic pressure.

3.7.1.8 Packings. Wherever possible, packings and gaskets which conform to MIL-G-551O or MIL-P-5516 as applicable and, if feasible, shall be installed in accordance with MIL-P-5514.

3.7.1.9 Valves. A bleed valve shall be provided to permit draining of the oil out of the test hose back Into the reservoir after completion of the tests.

3.7.2 Air components.

3.7.2.1 Air Filter. An air filter of 10 micron filtration shall be provided at the air inlet port. The air filter shall be equipped with a moisture trap, glass or plastic sight bowl, and a drain valve.

3.7.2.2 Air regulator, An adjustable air regulator valve shall be provided in the air inlet line to control the hydraulic outlet pressure between 300 and 15,000 psi.

3.7.2.3 Air lubricator. A lubricating device shall be installed in the inlet line between the regulator and the pump to supply lubrication 3 to 4 drops per minute to the actuated pump.

3.7.2.4 Air muffler. A suitable muffler shall be provided which will reduce the noise to a level not to exceed 80 decibels at a distance of 5 feet in any direction from the muffler.

3.7.2.5 Air shut-off valve. A shut-off valve shall be provided in the air line to the pump to shut off air flow to the pump.

3.8 Interchangeability. All parts having the same manufacturer’s part number shall be functionally and dimensionally interchangeable. The

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drawing number requirements of MIL-STD-1OO shall govern changes in the manufacturer's part numbers.

3.9 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.10 Finish, The reservoir, cover, and all united parts made of aluminum alloy shall be anodized in accordance with MIL-A-8625. All steel parts shall be cadmium plated in accordance with QQ-P-416 or shall be painted. If the gage cases are steel, they shall be cadmium plated or painted black, The surfaces to be painted shall be prepared by cleaning and sand or grit blasting. A coat of zinc chromate primer, MIL-P-8585, shall be applied in accordance with MIL-P-6808. The top coat consisting of 2 coats of lacquer, MIL-L-7178, or 2 coats of black enamel, MIL-E-7729, type III, shall then be applied.

3.11 Workmanship. Workmanship shall be first class in every respect, and free from defects that could affect appearance and satisfactory performance. Installation of equipment and accessories shall be in a thoroughly workmanlike manner. Particular attention shall be given to accuracy of dimensions, marking of parts and assemblies, soldering, painting, wiring, and alignment of parts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government, The Government reserves the right to perform any of the inspections set forth In the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of examinations and tests. Examinations and tests of the tester hose assemblies shall be classified as (1) first article. and (2) quality conformance.

4.2.1 First article. Unless otherwise specified in 6.2, the first article shall be examined and tested as specified in 4.3. Failure of the first article to pass any examination or test shall be cause for rejection.

4.2.2 Quality conformance. Quality conformance tests shall be based upon compliance of each of two samples selected at random from any lot

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of 10 testers, or any fraction thereof, manufactured under one contract, under essentially the same conditions. These two samples shall be subjected to the test procedures specified in 4.3.

4.3 Test condition.

4.3.1 First article. First article test shall consist of individual tests.

4.3.2 Individual tests. Each tester shall be subjected to the following tests as described under 4.5 quality conformance of this specification.

a. Examination of product
b. Functional tests
c. Static test
d. Gage test

4.4 Test methods.

4.4.1 Examination of product. Each tester shall be inspected to determine compliance with requirements specified herein with respect to parts, materials, finish, workmanship, dimensions, and markings.

4.4.2 Functional tests. Using a tester as the pressure source, a proof pressure test and a leakage test shall be conducted individually first on an MS28741-4-0120 hose assembly, and then on an MS28741-12-012 hose assembly. These tests shall also be conducted individually, first on an MS28759-4-0120 hose assembly and then on an MS28759-12-0120 hose assembly. The proof pressure and leakage pressures shall be used as indicated in the table below. (Tests shall be conducted in accordance with FED-STD-601, method 1011.) Strain, leakage at any point, failure of any component, or a reduction of more than 1% of specified test pressure shall be cause for rejection. MIL-H-83282 shall be used for all functional tests.

TABLE I

PROOF PRESSURE AND LEAKAGE TESTS

ASSEMBLY	PROOF PRESSURE - PSI	LEAKAGE PRESSURE - PSI	TIME APPLICATION OF PRESSURE
MS28741-4-0120	6000	8400	5 min.
MS28741-12-0120	3000	4200	5 min.
MS28759-4-0120	8000	11200	3 to 5 min.
MS28759-12-0120	6000	8400	3 to 5 min.

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4.4.3 Static test. The high pressure hydraulic system, with the exception of the gages, shall be subjected to a proof pressure of 1-1/2 times the maximum operating pressure for a period of 6 minutes. There shall be no leakage, evidence of strain, nor failure of any components. MIL-H-83282 shall be used for all static tests.

4.4.4 Gage test. Each gage shall be calibrated at full scale against a master gage or tester with an inherent accuracy of 1/4 of 1%. Each gage shall then be cycled from O to full scale to 0 ten tires. Each gage shall then be rechecked against the master gage or tester at full scale. A change from the initial calibration of more than 1% shall be cause for rejection.

4.5 Quality conformance,

4,5.1 Quality conformance test sample. The quality conformance test samples shall consist of 2 testers representative of the production equipment. They shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.5.2 Quality conformance test. Quality conformance test shall consist of all the tests described under 4.4. test methods.

4.6 Inspection of the preservation, packaging and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of Section 5, for the documents specified herein.

5. PREPARATION FOR DELIVERY.

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each tester shall be packaged Method IA-14 of MIL-P-116, subject to the preproduction inspection of MIL-P-116. The unit container shall conform to PPP-B-601, Grade A.

5.1.2 Level B. Each tester shall be packaged as specified for level A except that the unit container may conform to V3s, V4s, or V3c of PPP-B-636. Preproduction inspection shall be required.

5.1.3 Level C. Each tester shall be packaged in a manner to assure carrier acceptance and to prevent damage and deterioration during shipment from the supply source to the receiving activity for immediate use.

5.2 Packing. No packing required.

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5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The test unit covered by this specification is intended for conducting proof pressure and leakage tests on aircraft and missile hoses which conform to MIL-H-8788, MIL-H-8790, and MIL-H-8795.

6.2 Ordering data. Procurement document should specify the following:

a. Title, number, and date of this specification.

b. Whether first article approval is required (see 3.1 and 4.2.1).

Level of preservation and packaging and packing required (see 5.1).

d. Whether special marking for shipment is required (see 5.3).

Custodian:

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Preparing activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

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A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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